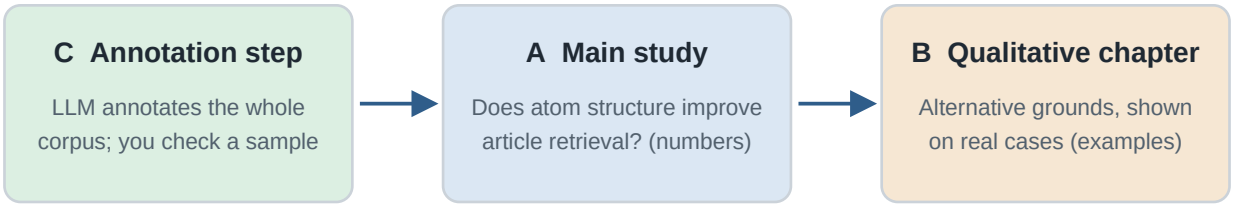
Thesis plan: finding the law below the article

Bachelor thesis, Computer Science. Building on the Article Relationship Model and on G-DSR (Louis, Van Dijck & Spanakis, EACL 2023). October 2026.

The idea in one paragraph

Laws are long, dense texts. Search systems usually treat a whole article as one block of text. This project splits each article into **atoms**: small pieces that each say one legal thing, such as "the care provider keeps a file" or "only if this is necessary for good care". Each atom is labelled with who acts, what they do, when, and which other articles it points to. The thesis asks one main question: **does this fine-grained structure help a computer find the right law for a real legal problem?**



One thesis, one corpus, one system: C builds the data, A measures it, B shows what it means for a lawyer.

Part	Role in the thesis	Type of result	Share of time
A Main study	Measure whether atom structure improves retrieval of Dutch statutory articles	Numbers: recall, ranking scores, ablations	~55%
C Annotation step	Produce enough annotated articles for A, and check how reliable the annotations are	Numbers: agreement between LLM and human	~25%
B Qualitative chapter	Show how the graph finds alternative legal grounds in a few real cases	Worked examples, discussion	~20%

Where the project stands today

There is a working pipeline: 80 annotated atoms from 30 real articles (Book 6 and the WGBO, Book 7), four ways of linking atoms (shared labels, similar labels, the legal hierarchy, explicit references), a scoring function, a graph walker that explains its steps, and a Streamlit app. On the current test ("given one atom, find the article it came from") it scores top-1 accuracy 0.48 and MRR 0.55. That test is useful for debugging, but it is not a task anyone needs. The thesis replaces it with a real one.

A. Main study: does atom structure improve article retrieval?

Research question. Given the facts of a real Dutch legal case, does adding atom-level structure to a retrieval system help it find the statutory articles the court actually relied on?

Why this question

G-DSR showed that giving a retriever the *structure above the article* (the tree of books, titles and sections) makes it better. It treated each article as a single node. This project adds the *structure inside the article*. G-DSR's authors list richer graphs, especially citation links between articles, as future work. The atom graph includes those links. So the thesis tests a natural next step of your supervisor's own paper.

The task in plain words

The input is a short description of what happened in a case, for example "the hospital lost the patient's file and the patient cannot prove the treatment went wrong". The output is a ranked list of articles. A good system puts the articles the judge cited near the top.

Where the test data comes from

- **Questions:** real judgments from rechtspraak.nl (free open-data API). Take medical-liability cases. The facts section becomes the question.
- **Correct answers:** the articles of the BW that the judgment cites. These can be pulled out of the text automatically, so no hand-labelling is needed.
- **Article collection:** the full WGBO (articles 7:446 to 7:468) plus the Book 6 articles it leans on (damages, tort, liability for others), about 60 to 80 articles. Text and hierarchy are downloaded from wetten.overheid.nl.
- **Size:** aim for 200 to 500 case-question pairs. Split them into a development set (for tuning) and a test set (touched once, at the end).

The systems to compare

Each system is a small step up from the previous one, so you can see exactly what each addition contributes.

#	System	What it does
1	BM25	Classic keyword search. The standard baseline.
2	Dense retrieval	A ready-made multilingual embedding model (for example BGE-M3 or multilingual-e5) compares meaning, not just words.
3	Dense + atom re-ranking	An LLM splits the question into atoms with the same labels. The top results of system 2 are re-ordered by how well their atoms match the question's atoms (your existing scoring function).
4	Dense + atom graph	Scores are spread over the atom graph, so an article that is cited by, or shares actors and acts with, a strong match also moves up. This is the G-DSR idea, built from your atom links.

How results are measured

- **Recall@k:** of the articles the judge cited, how many are in the top k (for example k = 5, 10, 20)? This is what G-DSR reports.
- **MRR and nDCG@10:** how high the first correct article is ranked, and how good the order of the whole list is.
- **Ablation:** switch off one link type at a time (actors, acts, references, hierarchy) to see which one does the work.
- **Significance:** a paired test (for example a permutation test) so a small gain is not mistaken for a real one.

What counts as a good outcome

Both answers are publishable. If atoms help, you show how much and which kind of link matters. If they do not help, you show where and why (for example: dense models already capture this information), which is just as useful to the lab. What matters is a fair, well-measured comparison.

C. Annotation step: can an LLM annotate law the way the lab does?

Question. How closely does an LLM follow the lab's 9-label annotation guidelines, compared with a human annotator?

Why it is needed

Study A needs 60 to 80 annotated articles. Doing all of them by hand would take too long. Also, every current annotation was drafted by an LLM and nobody has checked it yet. Without a check, every result in A rests on unverified data. Part C fixes both problems at once.

How to do it

- **Automate.** Write a prompt (or a small tool-calling pipeline) that turns an article into atoms and fills in the 9 labels, using the lab's guidelines and a few of your examples. Save the output in the existing JSON format.
- **Gold sample.** Annotate 15 to 20 articles yourself, by hand, without looking at the LLM output first. If possible, ask a second person (a law student or someone from the lab) to annotate the same ones.
- **Compare.** Measure agreement: how often the atom boundaries match, pre/post-condition accuracy, and F1 per label (actors, acts, temporal, and so on). Report human-vs-human agreement too, because that is the ceiling.
- **Link to A.** Run study A twice: once on the LLM annotations and once on the hand-checked ones (for the sample). If the scores are close, the LLM annotations are good enough for retrieval.

Ask your supervisor first

The lab's guidelines mention annotation work on EU legislation. If the lab already has human-annotated provisions, use them as extra gold data. That makes part C much stronger and saves you time.

B. Qualitative chapter: finding alternative legal grounds

Question. When a claim under one article fails, can the atom graph point a lawyer to other articles that could still work, and explain why?

This was the project's original motivation. It is hard to measure with numbers because good test data is rare, so it becomes a chapter of worked examples instead of a benchmark.

- **Pick 3 to 5 real cases** where a lawyer argued one article first and another as backup (in Dutch: *primair* and *subsidiair*). A good anchor is HR 17 December 2010 (ECLI:NL:HR:2010:BN6236): missing medical records (7:454) combined with a tort claim (6:162).
- **Run the graph walker** from the failing article and show its trace: which atoms it reached, through which link, and why.
- **Compare** what the walker found with what the court and the lawyers used. Did it find the backup article? Did it find nonsense? Why?
- **Discuss** what the explanations are worth. A lawyer can check every step, which a plain ranked list cannot offer.

Timeline (about 6 months, can stretch to 7 or shrink to 4)

Month	Work	Deliverable
1	Download statutes from wetten.overheid.nl. Build the rechtspraak.nl scraper and the article-citation extractor. Agree on the research question with your supervisor.	Article collection + first 100 case questions
2	**C:** LLM annotation pipeline. Hand-annotate the gold sample. Measure agreement.	Annotated corpus + agreement table
3	**A:** baselines (BM25, dense). Finish the dataset (200 to 500 questions) and freeze the test split.	Baseline results
4	**A:** atom re-ranking and atom graph systems. Tune on the development set only.	Systems 3 and 4 working
5	**A:** final test run, ablations, significance tests. **B:** case studies.	All results + case-study chapter draft
6	Write up. Feedback round with your supervisor. Final edits.	Thesis

If you only have 4 months: use 100 to 200 questions, drop system 4, and keep B to two cases. If you have 7: add a small user check where a law student rates the walker's explanations.

Risks and backup plans

Risk	Backup plan
Too few judgments cite WGBO articles clearly.	Widen to more of Book 6 and Book 7, or use the judgment summaries (inhoudsindicatie) as shorter questions.
LLM annotations are poor.	That is itself a result for C. Use the hand-checked subset for A, and report the gap.
Atoms give no gain over dense retrieval.	Still a valid finding. The ablation and error analysis explain why, and B shows where structure still helps (explanations).
Citing an article is not the same as it being relevant.	Hand-check a sample of 30 to 50 pairs and report how noisy the labels are.

Done already (October 2026)

- The synthetic "Lunar Medical Code" is removed from evaluation and kept only as an automated test (`tests/test_pipeline.py`). The honest numbers on real articles went up: top-1 0.36 to 0.48, MRR 0.42 to 0.55.
- The graph walker lives in `src/data/reasoning.py`, so the app, the notebook and future experiments share one version.

Questions for the next supervisor meeting

- Is the research question of study A the right one for the lab?
- Does the lab have human-annotated provisions (from the EU annotation work) that part C can use?
- Is there a law student or lab member who could annotate 15 to 20 articles as a second annotator?
- Note for the project description: G-DSR uses the hierarchy only. Citation links are named in its Limitations as future work, which this thesis adds.